Basic Concepts in CNS Pharmacology

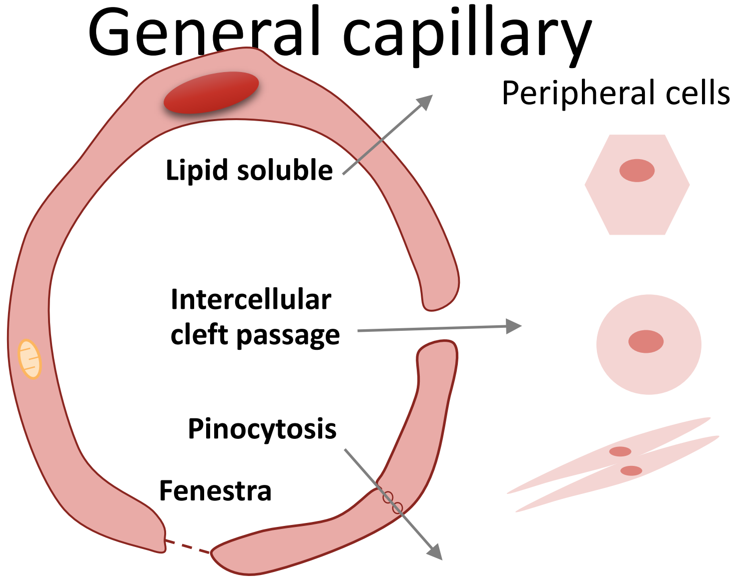
Pongsatorn Meesawatsom
Department of Pharmacology
Faculty of Pharmacy
Mahidol University

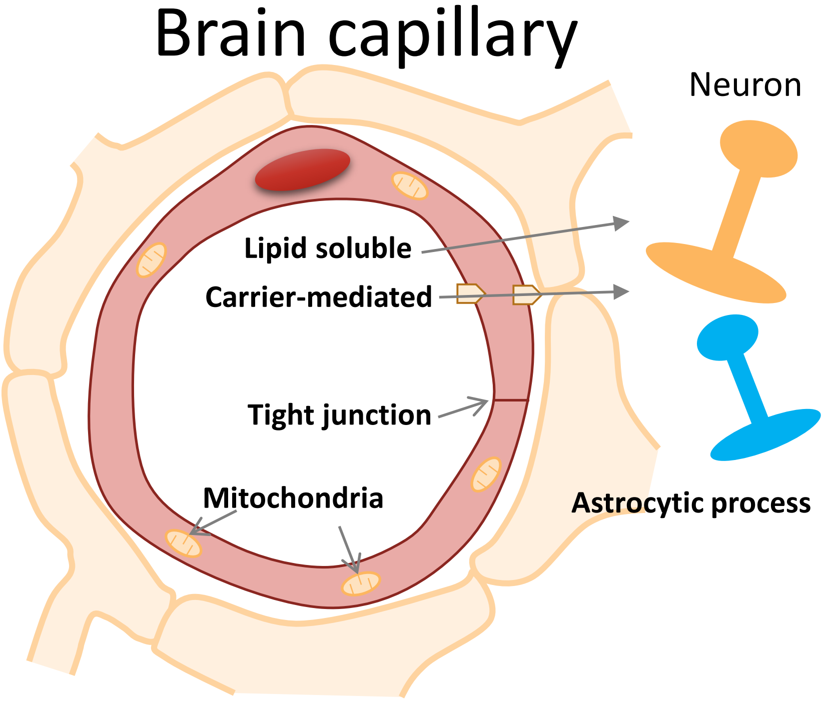
1. Blood-brain barrier (BBB)
2. Major drugs targets are modulations of the synaptic neurotransmission and neuronal excitability
3. Pharmacological effects of drugs acting in the CNS are the combination of effects at the PNS and CNS
4. Pharmacological effects are mediated by the action of drugs at certain receptors and in specific brain areas
5. Some CNS functions are controlled by many neurotransmitters
6. Some CNS drugs are promiscuous and have different effects on receptors

CNS Pharmacology: Concept 1
Blood-brain barrier (BBB)

Blood-brain barrier (BBB)

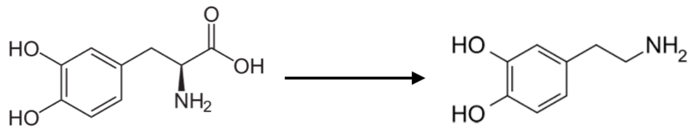
- Brain is “protected” by a specialized system of capillary endothelial cells known as the BBB
- Unique properties
  - Endothelium connected with tight junction
  - Mitochondria-riched endothelium
  - Surrounded by astrocytic process





Major properties of drugs that can pass BBB

- Sufficient lipophilicity
  - Diazepam, barbiturates, opioids
- Small molecule
  - Halothane, alcohol, nitrous oxide
- Transported via specific transporters
  - Large neutral amino acids transporter (LNAAT or LAT)
    - levodopa, L-amino acid, gabapentin, pregabalin

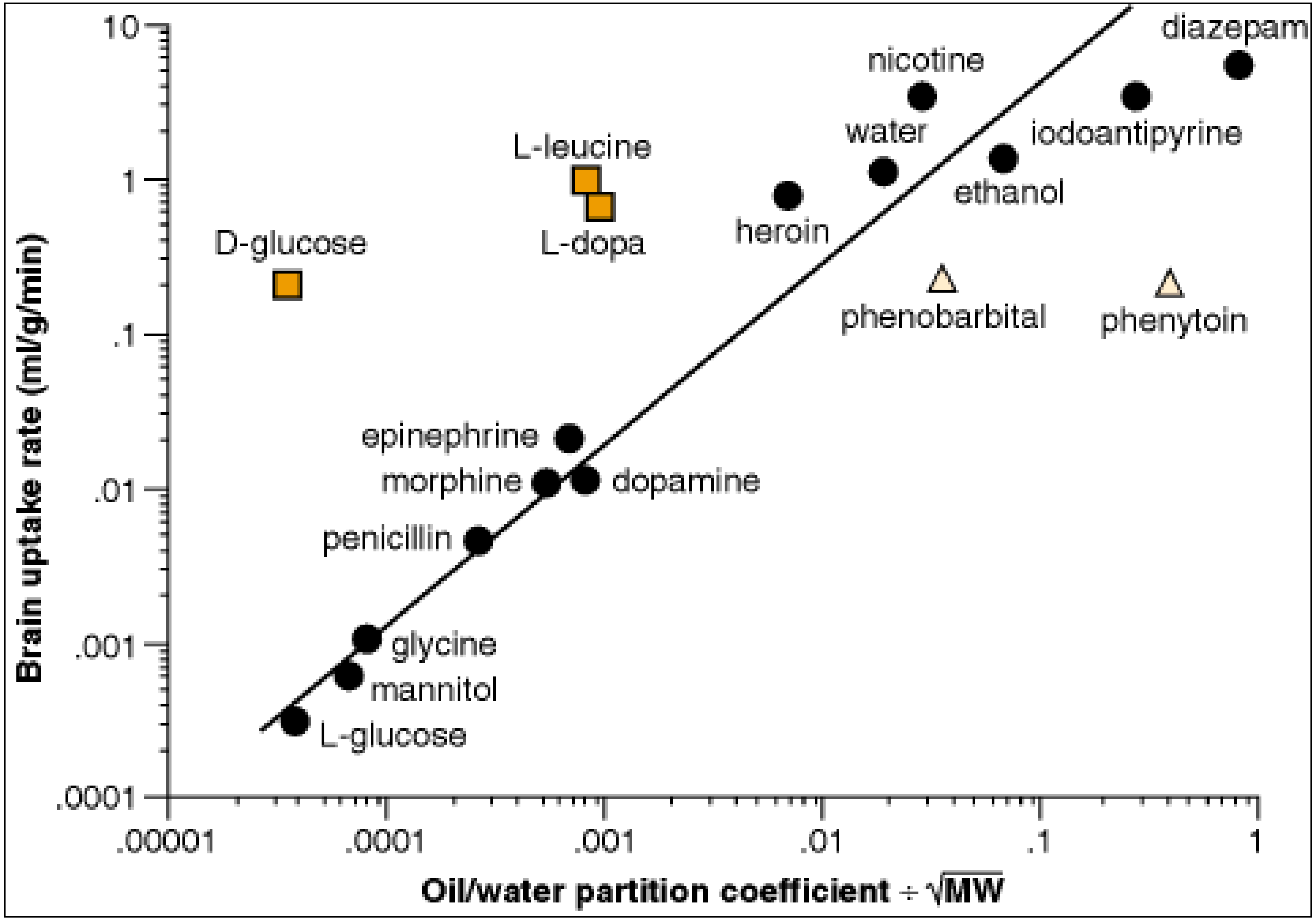


L-DOPA Dopamine

Efflux transporters – P-glycoprotein (Pgp)

Some drugs are substrates of Pgp e.g., phenobarbital, phenytoin

Relation of lipophilicity and brain uptake rate of some CNS drugs



Brain areas out of BBB

- Pituitary gland
- Chemoreceptor trigger zone (CTZ)
- Arcuate nucleus of hypothalamus (ARH)

CNS Pharmacology: Concept 2
Major drugs targets are modulations of the synaptic neurotransmission and neuronal excitability

General steps of chemical synaptic neurotransmission by neurotransmitter (NT)

- Synthesis
- Storage
- Release
  - Intracellular Ca^{2+} dependent process
- Receptor activation
  - Post-synaptic receptors
  - Presynaptic receptors
- Synaptic turnover
  - Reuptake
  - Metabolism

Neuronal excitability can control the release of neurotransmitters

Voltage-gated Na^{+} channels

Voltage-gated Ca^{2+} channels

Voltage-gated K^{+} channels

Neurotransmitter	Rate-limiting step in synthesis	Removal/inactivation/ turnover mechanism
Acetylcholine (ACh)	Choline acyltransferase (CAT)	Acetylcholinesterase (AChE)
Norepinephrine (NE)	Tyrosine hydroxylase	Norepinephrine transporter (NET) MAO-A, COMT
Dopamine (DA)	Tyrosine hydroxylase	Dopamine transporter (DAT) MAO-B, COMT
Serotonin (5-HT)	Tryptophan hydroxylase	Serotonin transporter (5-HTT or SERT) MAO-A, MAO-B
Histamine	Histidine decarboxylase	MAO, histamine methyl transferase and possibly transporter
GABA	Glutamic acid decarboxylase (GAD)	GABA transporters (GATs) neurons and glia
Glutamate	Glutaminase	Several transporters on neurons and glia

Presynaptic receptor concepts

Autoreceptor

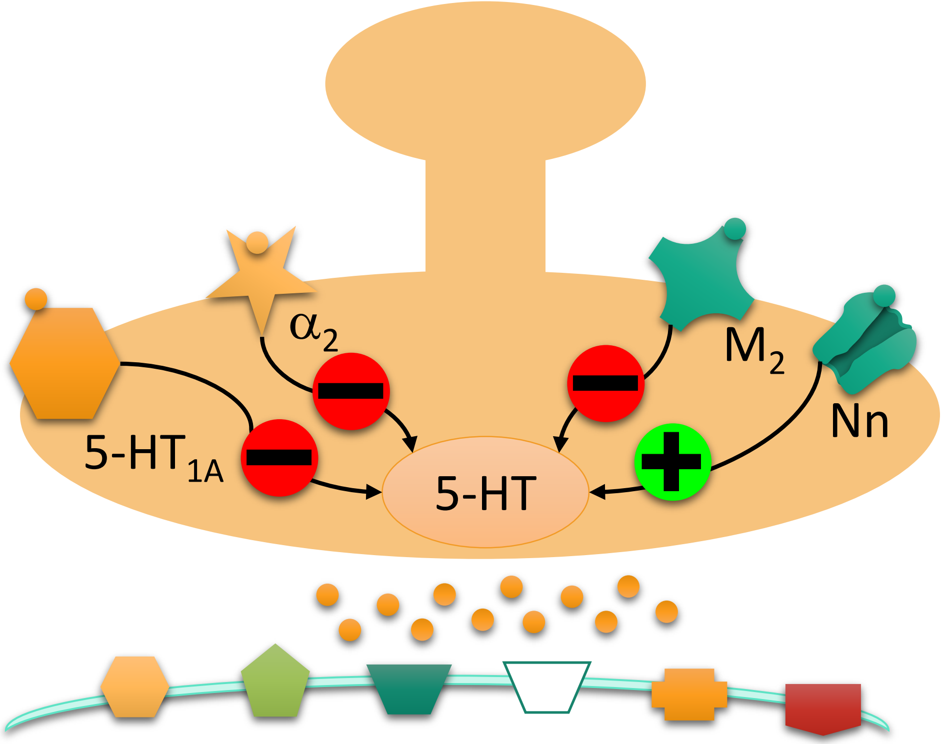
The presynaptic receptor is for the same NT released by the presynaptic neuron

Heteroreceptor

The presynaptic receptor is for the different NT that released by the presynaptic neuron

Presynaptic receptor activation may either enhance or inhibit the NT release.

Receptor	Signaling
$5\text{-}HT_{1A}$	Gi
α_2	GI
Nn(nAChR)	Ca^{2+}, Na^{+}
M_2	Gi



CNS Pharmacology: Concept 3 Pharmacological effects of drugs acting in the CNS are the combination of effects at the PNS and CNS

Diphenhydramine

Antihistamine effects

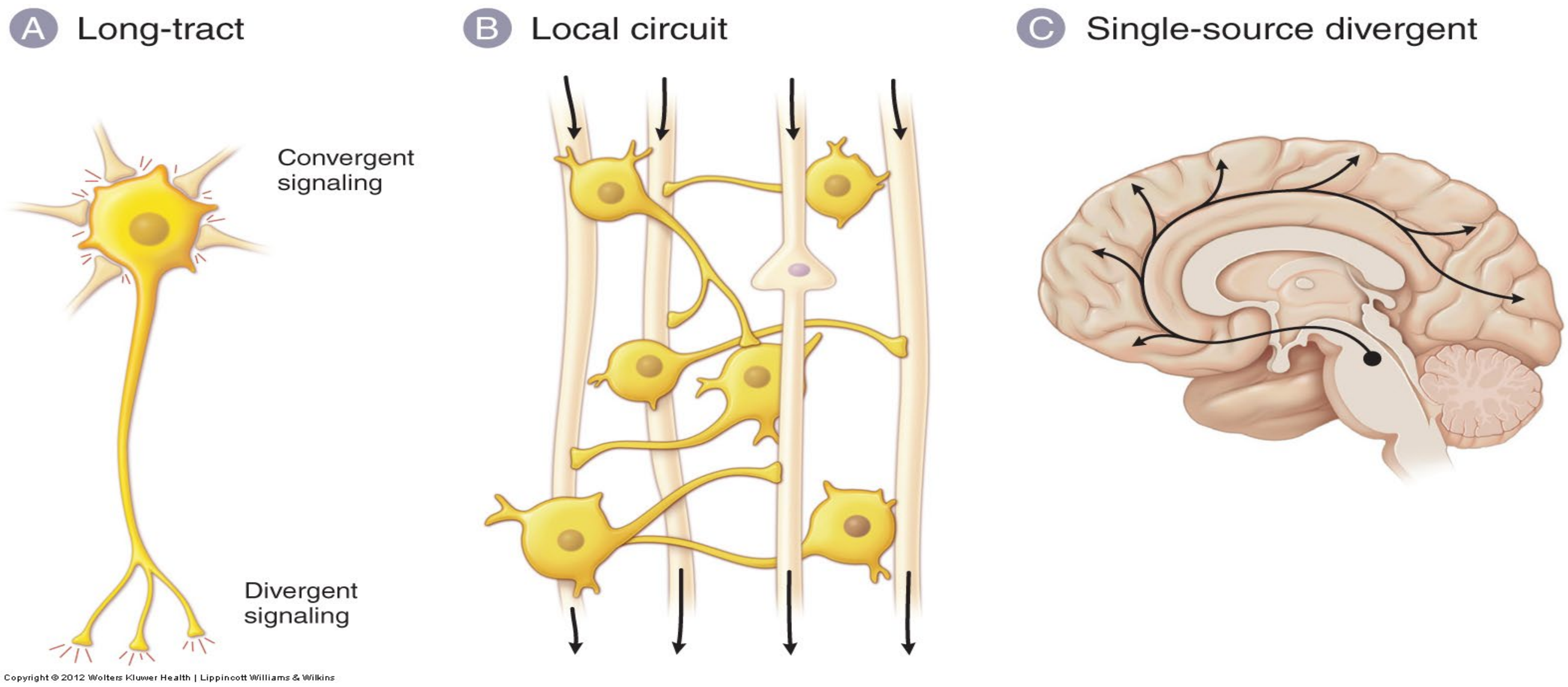
- Peripheral
  - Therapeutic effects
- Central
  - Drowsiness, somnolence
  - Increase appetite

Anticholinergic effects

- Peripheral
- Central
  - Drowsiness, somnolence
  - Amnesia
  - Suppress REM sleep
  - Impaired cognition
  - Antiparkinsonism (mild)

CNS Pharmacology: Concept 4
Pharmacological effects are mediated by the action of drugs at certain receptors and in specific brain areas

Neuronal organization in CNS



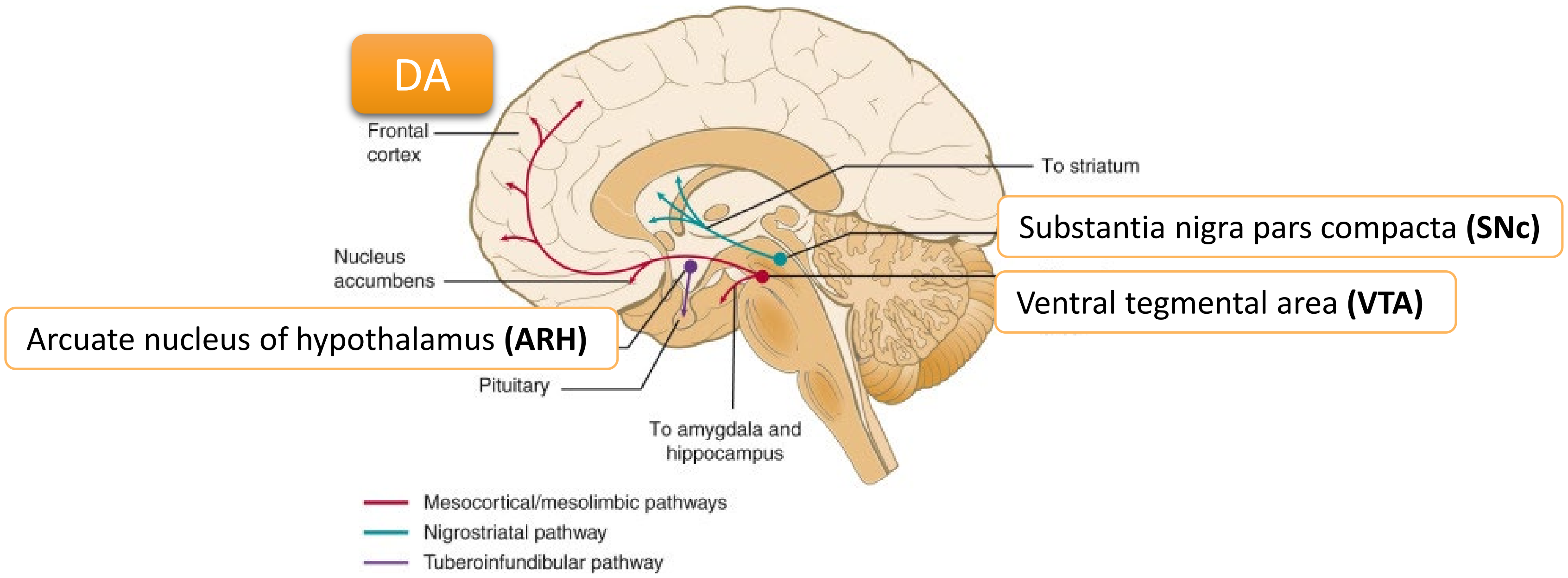
Corticospinal tracts

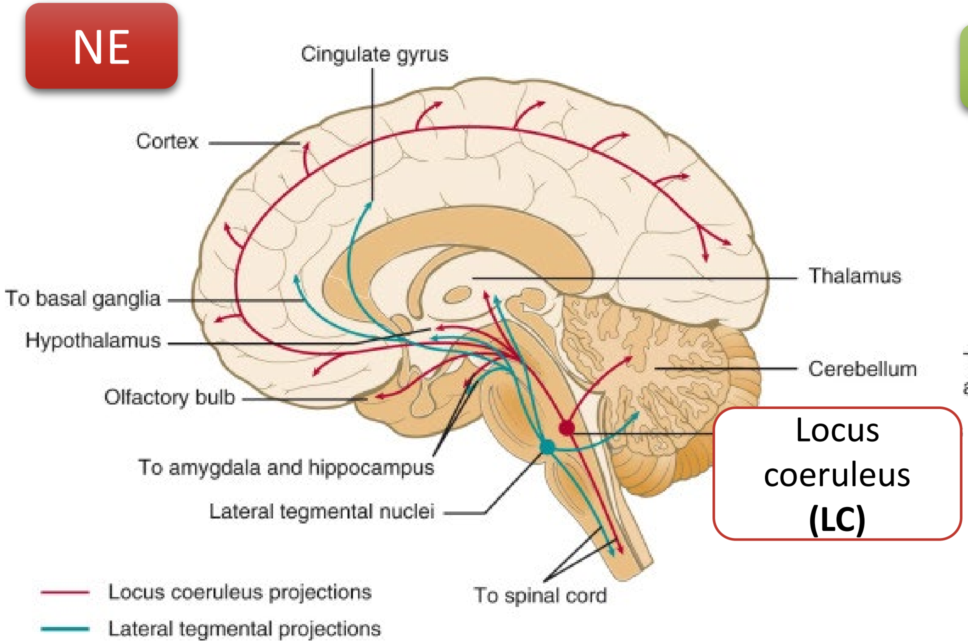
GABA or glutamate interneurons (widely distributed throughout the CNS)

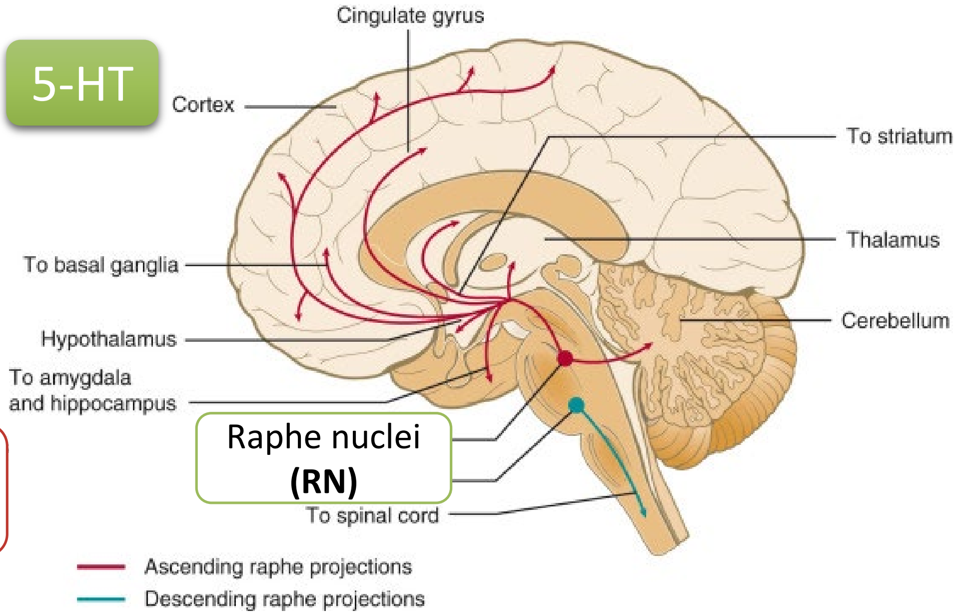
Glycine interneuron (Renshaw cell) in spinal cords

ACh interneurons in striatum

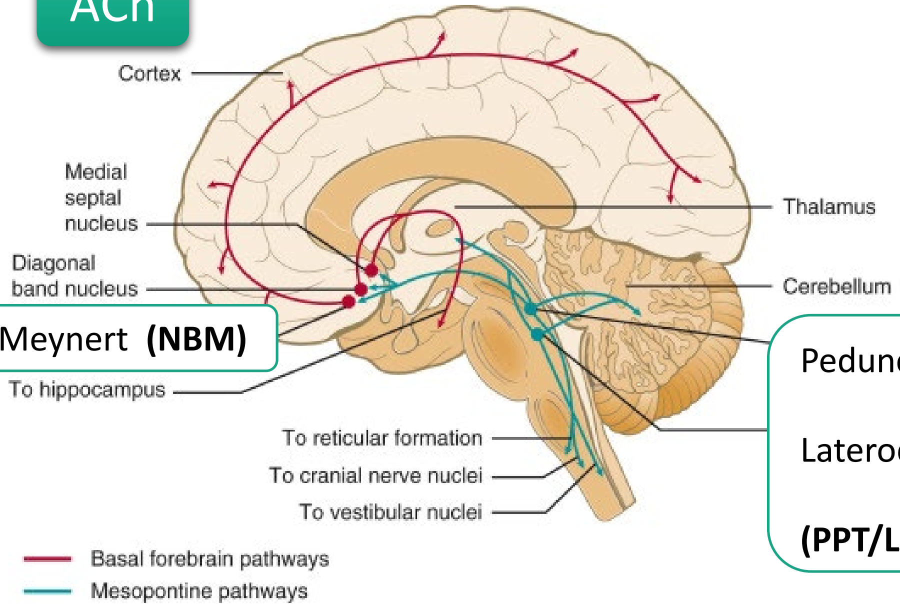
Monoaminergic nuclei


DA
Frontal cortex
Nucleus accumbens
Arcuate nucleus of hypothalamus (ARH)
Pituitary
To amygdala and hippocampus
To striatum
Substantia nigra pars compacta (SNc)
Ventral tegmental area (VTA)
Mesocortical/mesolimbic pathways
Nigrostriatal pathway
Tuberoinfundibular pathway


NE
Cingulate gyrus
Cortex
To basal ganglia
Hypothalamus
Olfactory bulb
To amygdala and hippocampus
Lateral tegmental nuclei
Thalamus
Cerebellum
Locus coeruleus (LC)
To spinal cord
Locus coeruleus projections
Lateral tegmental projections


5-HT
Cingulate gyrus
Cortex
To striatum
Thalamus
To basal ganglia
Hypothalamus
To amygdala and hippocampus
Cerebellum
Raphe nuclei (RN)
To spinal cord
Ascending raphe projections
Descending raphe projections

ACh



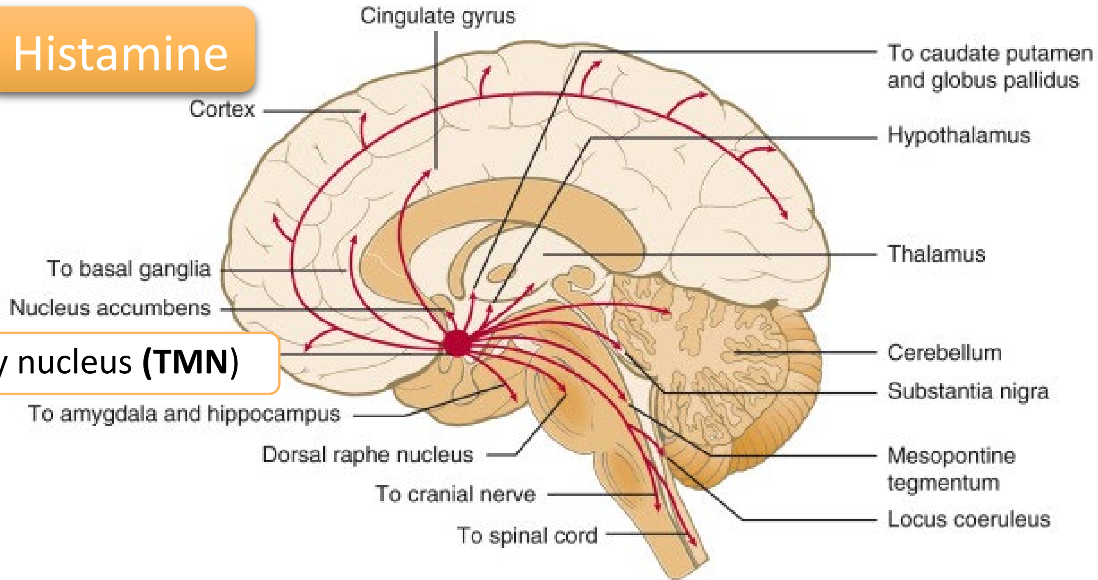
Nucleus basalis of Meynert **(NBM)**

Pedunculopontine

Laterodorsal tegmental nucleus

(PPT/LDT)

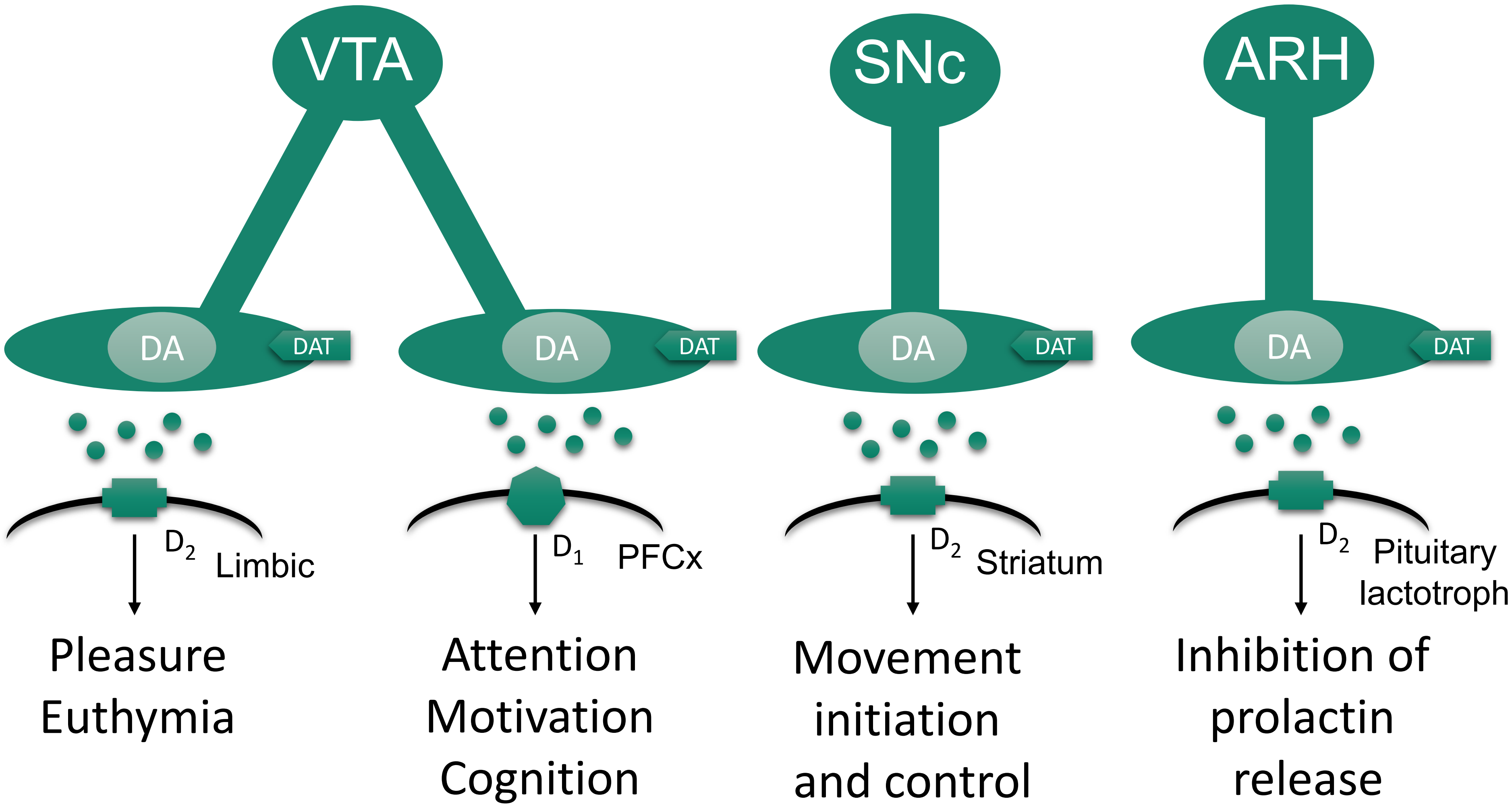
Histamine

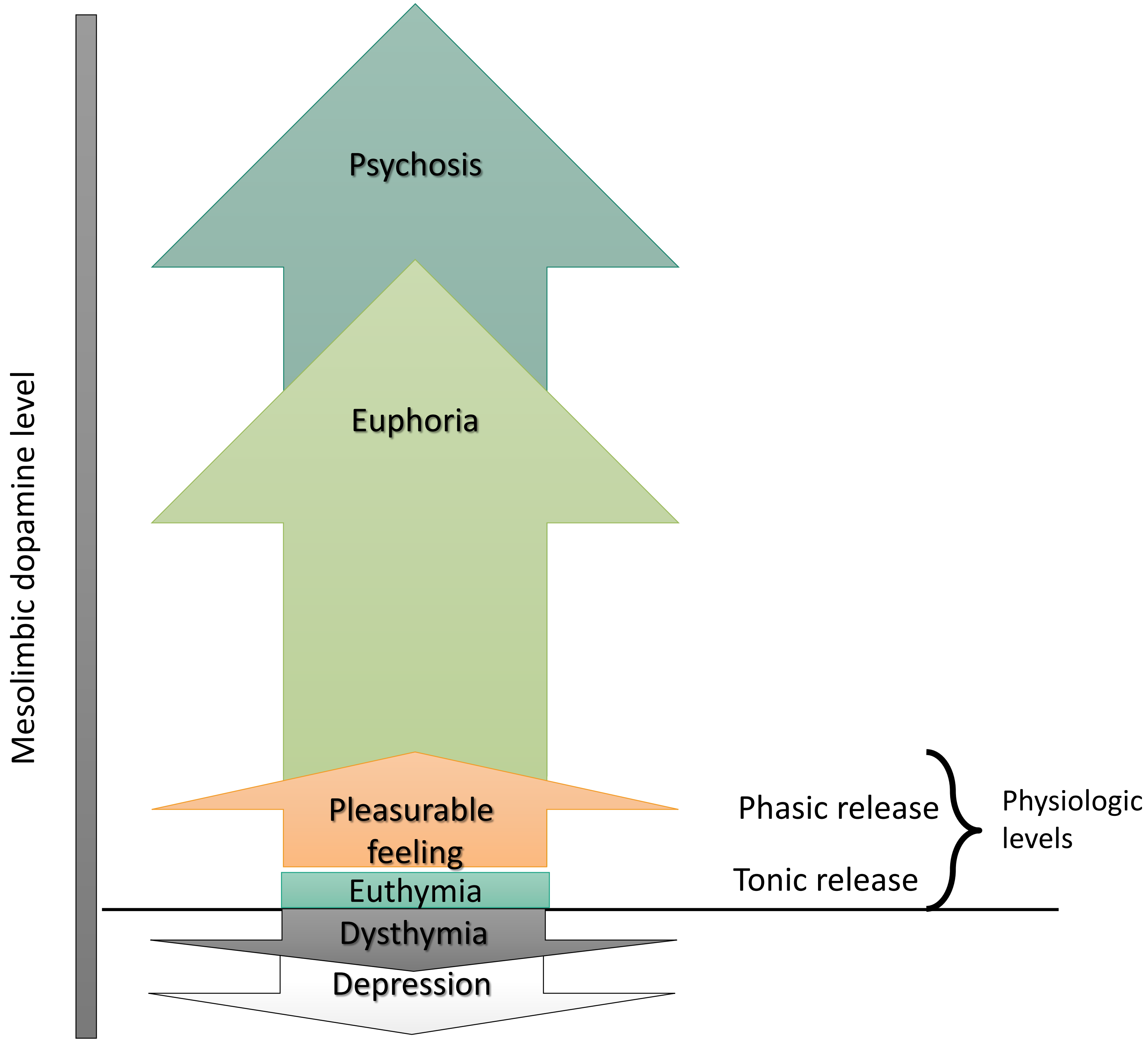


Tuberomammillary nucleus **(TMN)**

Neurotransmitter	Principle nuclei	Rate-limiting step in synthesis	Removal/inactivation/ turnover mechanism
Acetylcholine (ACh)	Nucleus basalis of Meynert (NBM) Pedunculopontine and laterodorsal tegmental nucleus (PPT/LDT)	Choline acyltransferase (CAT)	Acetylcholinesterase (AChE)
Norepinephrine (NE)	Locus coeruleus (LC)	Tyrosine hydroxylase	Norepinephrine transporter (NET), MAO-A, COMT
Dopamine (DA)	Substantia nigra pars compacta (SNc), ventral tegmental area (VTA), Arcuate nucleus of hypothalamus (ARH)	Tyrosine hydroxylase	Dopamine transporter (DAT), MAO-B, COMT
Serotonin (5-HT)	Raphe nuclei (RN)	Tryptophan hydroxylase	Serotonin transporter (5-HTT or SERT), MAO-A, B
Histamine	Tuberomammillary nucleus (TMN)	Histidine decarboxylase	MAO, histamine methyl transferase and possibly transporter
GABA	Widely distributed in interneurons	Glutamic acid decarboxylase (GAD)	GABA transporters (GATs) neurons and glia
Glycine	Widely distributed in interneurons	Phosphoserine	Glycine transporters (GLYTs) neurons and glia
Glutamate	Widely distributed in interneurons	Glutaminase	Several transporters on neurons and glia
ATP	Widely distributed as a co-transmitter	Mitochondrial oxidative phosphorylation; glycolysis	Hydrolysis to AMP and adenosine
Endocannabinoids (anadamide, 2-arachidonoyl glycerol)	Widely distributed as a retrograde transmitter	N-acylphosphatidylethanolamine (NAPE)-specific phospholipase D (PLD) (NAPE-PLD DAG-lipase (DAGL)	Fatty acid amidohydrolase (FAAH) Monoacylglycerol lipase (MAGL)
Neuropeptides; orexin, enkephalin, endorphin	Various	Synthesis and transport	Proteases

Dopaminergic transmission: Physiologic states

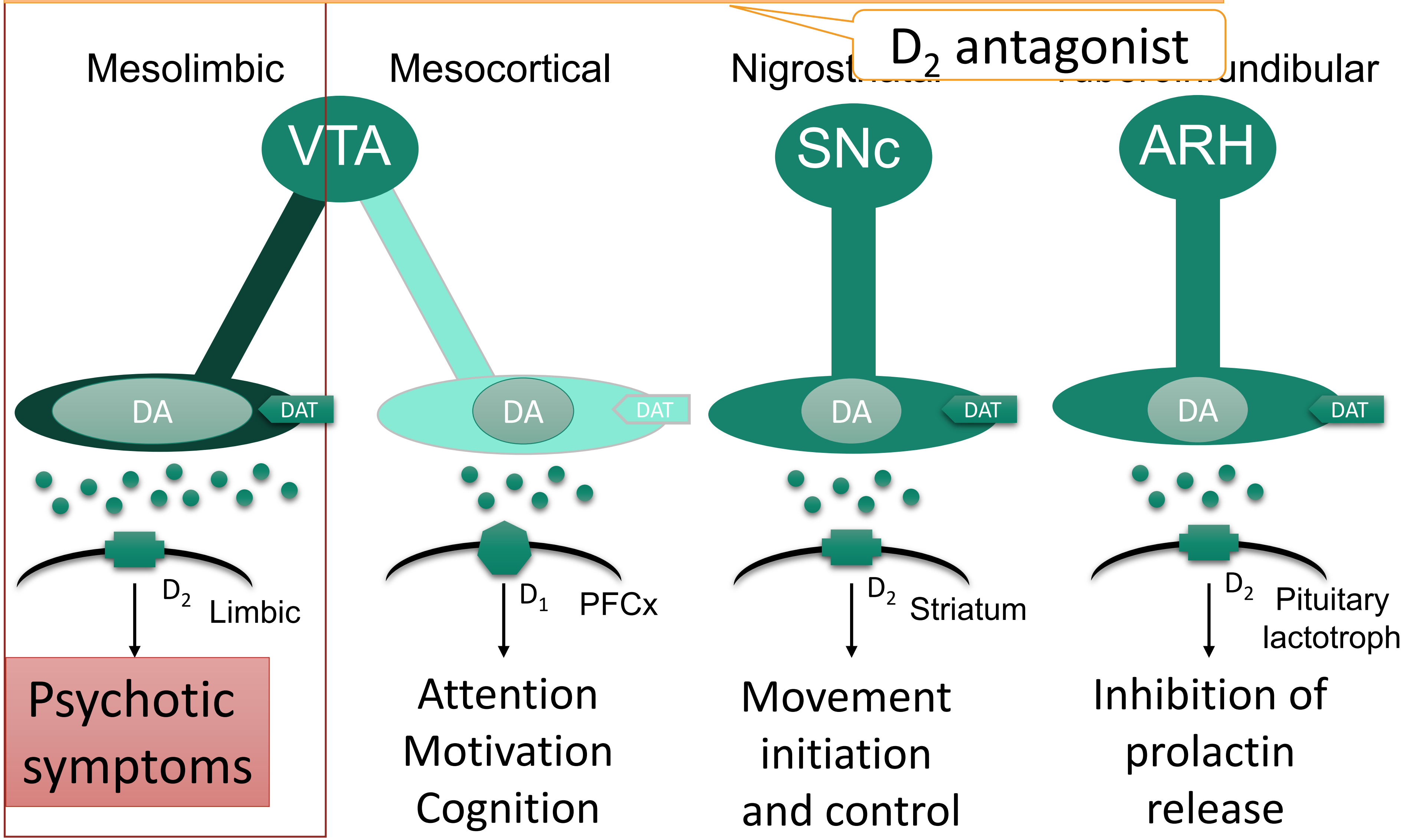



Mesolimbic dopamine level
Psychosis
Euphoria
Pleasurable feeling
Euthymia
Dysthymia
Depression
Phasic release
Tonic release
Physiologic levels

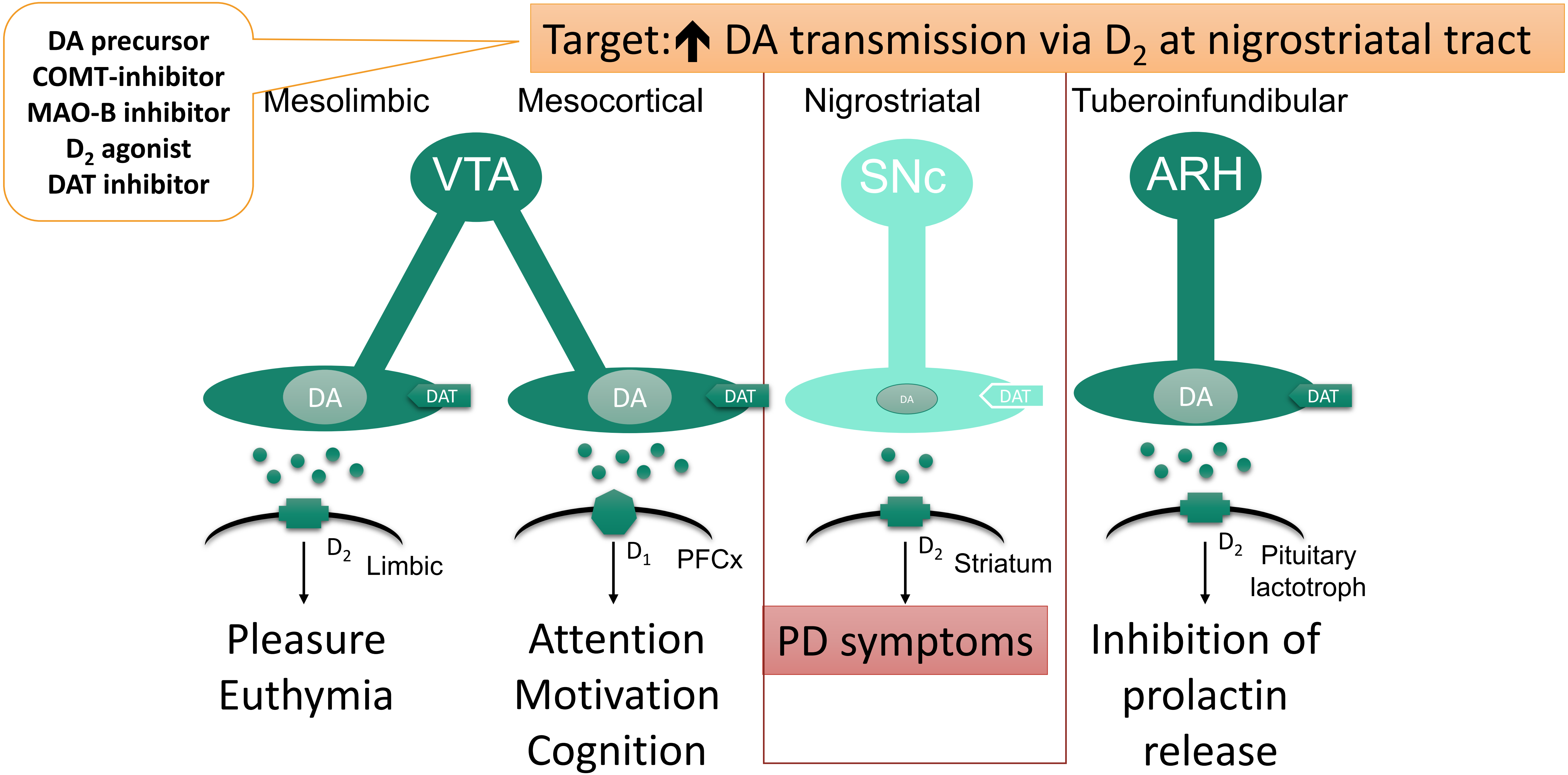
Dopaminergic transmission: Schizophrenia

Target: ↓ DA transmission via D_2 at mesolimbic tract

D_2 antagonist



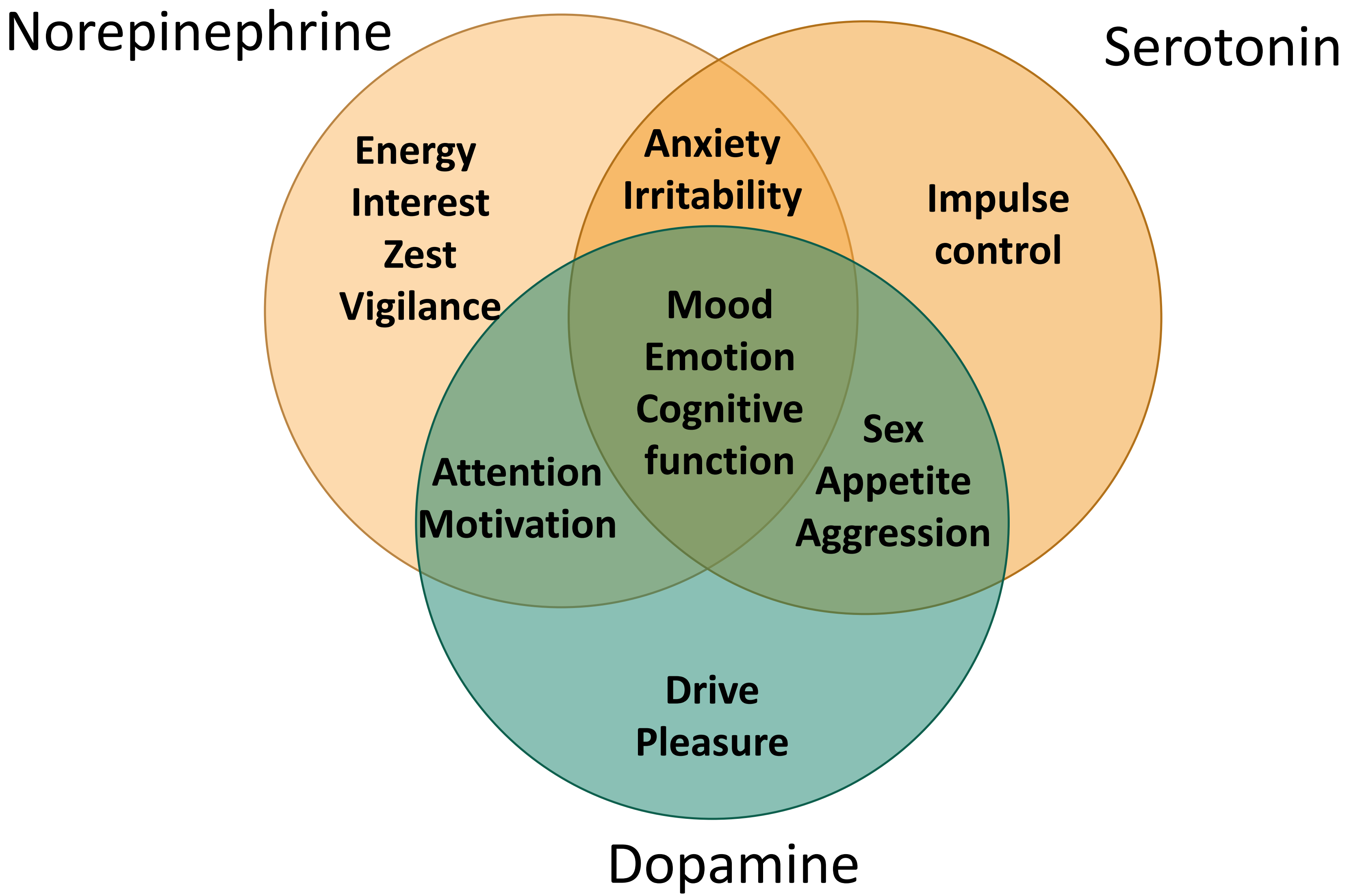
Dopaminergic transmission: Parkinson's disease (PD)



Nucleus and neurotransmitter	*Target innervation*	*Receptor at the target*	*Physiological responses*
RN: 5-HT	Prefrontal cortex	5-$HT_{1A, 2A, 2C, 6, 7}$	Executive function, cognition, impulse control
	Hippocampus	5-$HT_{1A, 2A, 2C, 6, 7}$	Euthymia, mood stabilization
	Amygdala	5-HT_{1A}	Anxiolysis
	Dorsal horn of spinal cord	5-HT_{1A}	Inhibit pain transmission
	Axon of VTA projecting to hippocampus and prefrontal cortex	5-$HT_{2A, 2C}$	Inhibit dopamine release from VTA
	Axon of SNc projecting to striatum	5-$HT_{2A, 2C}$	Inhibit dopamine release from SNc
	Axon of ARH projecting to pituitary lactotroph	5-HT_{2A}	Inhibit dopamine release from ARH
	Ventromedial hypothalamus	5-$HT_{2A, 2C}$	Promote satiety
	Reticular activating system and thalamic relay neuron	Various	Promote awake
	PPT/LDT	5-$HT_{2A, 2C}$	inhibit ACh release from PPT/DLT Inhibit REM sleep
	CTZ	5-HT_3	Nausea, vomiting

CNS Pharmacology: Concept 5
Some CNS functions are controlled by many neurotransmitters

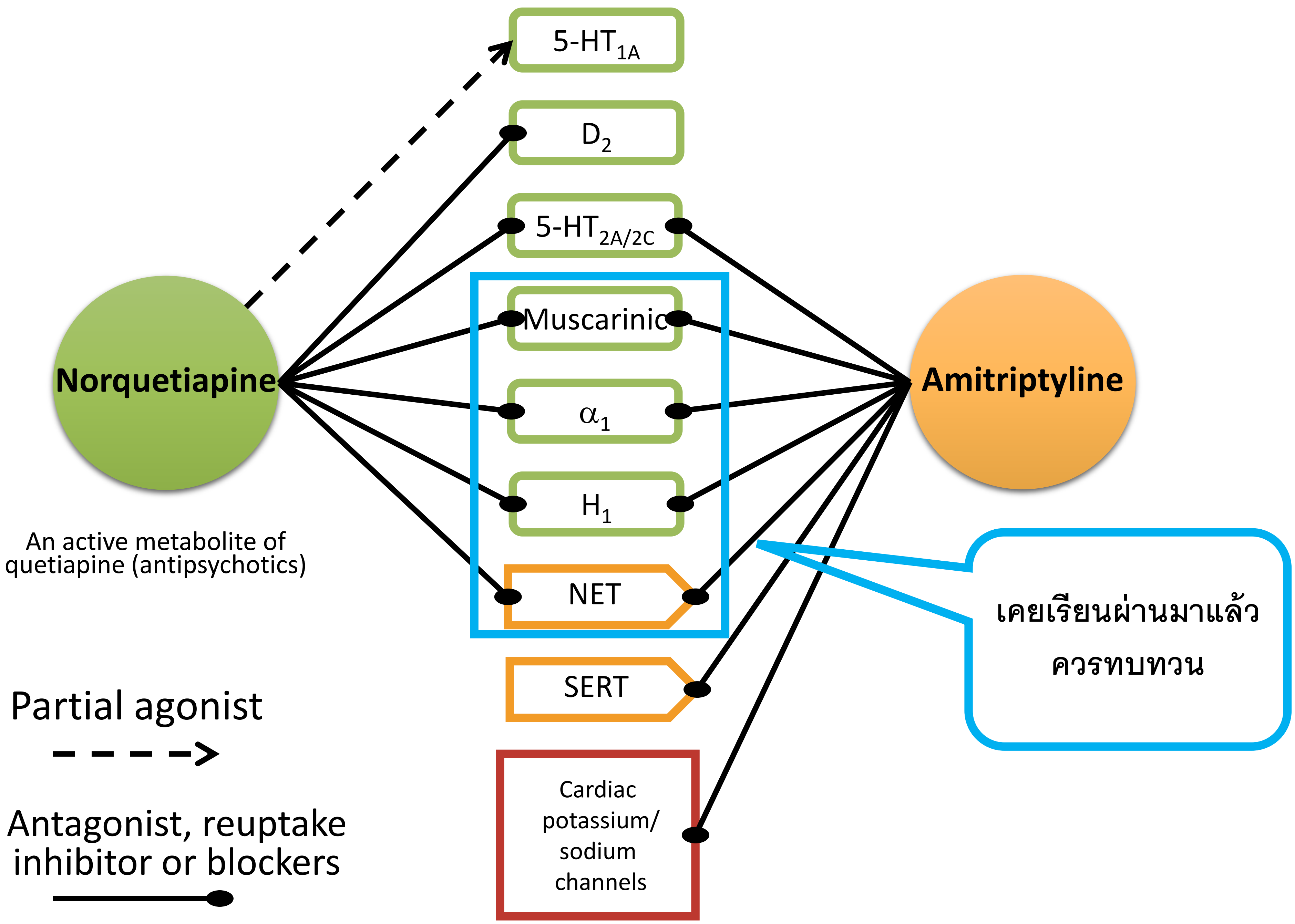
Monoamine neurotransmitters and their possible influence on psychopathology



Examples of CNS functions controlled by multiple neurotransmitters

- Sleep-wake cycle
  - NE, ACh, 5-HT, histamine, orexin
  - Melatonin, GABA
- Appetite feeding behavior
  - DA, NE, 5-HT, histamine
  - Orexigenic and anorexigenic neuropeptides
- Movement
  - DA, ACh, glutamate, GABA

CNS Pharmacology: Concept 6
Some CNS drugs are promiscuous and have different effects on receptors



5-HT$_{1A}$
D$_2$
5-HT$_{2A/2C}$
Muscarinic
α_1
H$_1$
NET
SERT
Cardiac potassium/ sodium channels
Norquetiapine
Amitriptyline
An active metabolite of quetiapine (antipsychotics)
Partial agonist
Antagonist, reuptake inhibitor or blockers
เคยเรียนผ่านมาแล้ว
ควรทบทวน